Structure of the EML completeness proof

An expository tour of grammars, combinators, and bridges

Abstract

We give a high-level account of how the *completeness theorem — every elementary function is expressible as a finite expression in the binary operator* $\mathrm{eml}(a, b) = \exp(a) - \log(b)$*, paired with the constant* 1 — is proved. The proof rests on a layered architecture: three inductive grammars (F36, EL, EML) connected by two compilers; a catalogue of *derived combinators* that recover the usual repertoire ($+$, $-$, $\cdot$, $/$, exp, log, $\sqrt{\ }$, ...) as fixed-shape sub-trees of the eml syntax; a single structural correctness theorem for the compiler; and a small zoo of *Euler bridges* from the complex EML grammar back to the real elementary functions. We trace one paper claim end-to-end as a worked example, and finish with the scoreboard summarising what is sealed. A closing addendum (§13) records the post- submission widening to full real-domain trig (Path C′), the cleanup of the Sheffer companion scaffolding (Plan A), and partial progress on per-primitive completeness for the paper's Sheffer cousins (Plans D and E).

Contents

1 The headline theorem

The paper's claim, stripped to bare mathematical content, is the following.

Theorem 1 (Headline)**.** *Let* $\mathsf{F36}$ *denote the catalogue of* 36 *paper primitives — atoms (*$\mathbf{1}$, π, e, -1, 2, $\frac{1}{2}$ *and the input variable), the eight standard real unaries, the six hyperbolic functions, the six trigonometric and inverse-trigonometric functions, and the eight binary operations. Let* $\mathcal{E}$ *be the single-binary-operator grammar*

$$\mathcal{E} \ni T ::= \mathbf{1} \mid \mathbf{x}_n \mid \mathrm{eml}(T,T), \qquad \mathrm{eml}(a,b) := \exp(a) - \log(b).$$

Then for each $f \in \mathsf{F36}$ *there is a finite expression* $\mathsf{T}_f \in \mathcal{E}$ *(or in its complex variant* $\mathcal{E}_{\mathbb{C}}$*) such that on a non-empty open subset of* f*'s natural domain, the partial evaluation of* T_f *yields a value whose appropriate projection* $\Re$ *or* $\Im$ *equals* $f(x)$*.*

The hedge *non-empty open subset* is honest. Three measure-zero points — $\sqrt{0}$, $\mathrm{arcosh}(1)$, $\mathrm{hypot}(0,0)$ — lie outside the natural construction; a few trig primitives are sealed on strict-but-symmetric subdomains rather than on the full real line. The bulk of the result, however, is a clean structural completeness, and the architecture below explains how the bulk is obtained.

The aim of this report is to explain *why* Theorem 1 is true: what shape the proof has, what each architectural layer contributes, and where the interesting ideas live.

2 Three languages, two compilers

The proof reduces the problem $f \rightsquigarrow \mathsf{T}_f$ through a three-language pipeline. The picture is

$$\begin{tikzcd} \boxed{\mathsf{F36}} \arrow[r, "\text{translate?}"] \arrow[drr, dashed, bend right, "\text{paper claim}"'] & \boxed{\mathsf{EL}} \arrow[r, "\text{compile}"] & \boxed{\mathsf{EML}} \arrow[d, hook, "\iota"] \\ & & \boxed{\mathsf{EML}_{\mathbb{C}}} \end{tikzcd}$$

The four boxes are inductive types; the three arrows are total functions; the dashed arrow names the public statement we want to verify.

$\mathsf{F36}$. The paper's source language. It has 36 named constructors and a partial denotation: each primitive returns $\perp$ outside its natural mathematical domain (e.g. log on ≤ 0).

EL. An intermediate language whose alphabet is *elementary*: atoms ($\mathbf{1}$, $\mathbf{x}_n$, named constants), the unaries $\{\exp, \log, \mathrm{neg}, \mathrm{halve}, \mathrm{sq}, \sqrt{\ }, \mathrm{inv}\}$, and the binaries $\{+, -, \cdot, /, \hat{}, \log_b, \mathrm{avg}, \mathrm{hypot}\}$. The translation $\mathsf{F36} \to \mathsf{EL}$ expands non-elementary names through standard identities: e.g.

$$\begin{aligned} \sinh(a) &\mapsto \tfrac{1}{2}\big(\exp(a) - \exp(-a)\big), & \tanh(a) &\mapsto \sinh(a)/\cosh(a), \\ \mathrm{arsinh}(a) &\mapsto \log\big(a + \sqrt{a^2+1}\big), & \mathrm{artanh}(a) &\mapsto \tfrac{1}{2}\big(\log(1+a) - \log(1-a)\big). \end{aligned}$$

The language EL is small enough to compile, but rich enough to absorb every named primitive of $\mathsf{F36}$.

EML. The pure single-operator grammar of Theorem 1. The compiler $\mathsf{EL} \to \mathsf{EML}$ is the heart of the proof.

$\mathsf{EML}_{\mathbb{C}}$**.** The complex-coefficient version of the EML grammar, with the same syntax but complex evaluation. Trig and the constants π, i require Euler's formula and therefore complex intermediates; $\mathsf{EML}_{\mathbb{C}}$ is where they live. The embedding ι is a *homomorphism*: it does nothing on syntax (every constructor maps to itself) and changes only the evaluation rule.

The architectural payoff is two short statements.

Theorem 2 (Real-fragment compiler)**.** *Suppose* $f \in \mathsf{F36}$ *satisfies* $\mathrm{translate?}(f) = \mathrm{some}\, e$ *with* $e \in \mathsf{EL}$*. Then for every environment* $\mathrm{env} : \mathbb{N} \to \mathbb{R}$ *and every value* $v \in \mathbb{R}$*,*

$$f.\mathrm{eval?}(\mathrm{env}) = \mathrm{some}\, v \implies \mathrm{compile}(e).\mathrm{eval?}(\mathrm{env}) = \mathrm{some}\, v.$$

Theorem 3 (Complex bridges)**.** *For each primitive in* $\{\pi, i, \cos, \sin, \tan, \arctan, \arccos, \arcsin\}$*, there is a finite witness* $\mathsf{C}_f \in \mathcal{E}_{\mathbb{C}}$ *and a projection* $\Pi \in \{\Re, \Im\}$ *such that, on a stated open subdomain of* f*'s natural domain,*

$$\mathsf{C}_f \downarrow v \quad \textit{with} \quad \Pi(v) = f(x).$$

The remaining sections open each box.

3 The kernel: EML grammar and partial evaluation

Definition 4 (EML grammar)**.** The set $\mathcal{E}$ of EML expressions is the smallest set generated by

$$\mathcal{E} \ni T ::= \mathbf{1} \mid \mathbf{x}_n \mid \mathrm{eml}(T, T),$$

where $n \in \mathbb{N}$. The complex variant $\mathcal{E}_{\mathbb{C}}$ has the same syntax; only the evaluation rule changes.

Definition 5 (Partial evaluation)**.** Fix an environment $\mathrm{env} : \mathbb{N} \to \mathbb{C}$. Define a partial function $\downarrow$ on $\mathcal{E}_{\mathbb{C}}$ by

$$\mathbf{1} \downarrow 1, \qquad \mathbf{x}_n \downarrow \mathrm{env}(n),$$

$$\mathrm{eml}(a, b) \downarrow \exp(\alpha) - \log(\beta) \quad \text{when} \quad a \downarrow \alpha,\ b \downarrow \beta,\ \beta \neq 0,$$

and undefined otherwise. The same rule with $\mathrm{env} : \mathbb{N} \to \mathbb{R}$ defines the real-evaluation variant on $\mathcal{E}$, where additionally the side condition $\beta > 0$ is required (so that $\log \beta \in \mathbb{R}$).

Why partial. A total evaluation would force a junk value at the boundary, e.g. the convention $\log 0 = 0$. Bridge theorems would then have to dance around the points where this convention disagrees with the natural mathematical answer. Partial evaluation lets us state, cleanly, "the witness has no value at the boundary, and the paper claim is stated on a subset that excludes it" — which is also what the paper does.

4 The combinator algebra

The single binary operator eml is too low-level to write useful witnesses with directly. The proof accumulates a catalogue of *derived combinators*, each defined as a fixed-shape sub-tree of $\mathcal{E}$ (or $\mathcal{E}_{\mathbb{C}}$), each accompanied by a *closure lemma* stating the conditions under which the combinator evaluates correctly.

Definition 6 (Catalogue)**.** The catalogue contains the following derived combinators. Each row gives the name, the value computed under the closure conditions, and the size of the underlying $\mathcal{E}$-tree.

Combinator	Computes (with appropriate side conditions)	Tree size
$\widehat{\exp}(T)$	$\exp(\alpha)$	2
$\widehat{\log}(T)$	$\log(\alpha)$, principal branch	4
$A \mathbin{\widehat{+}} B$	$\alpha + \beta$	11
$A \mathbin{\widehat{-}} B$	$\alpha - \beta$	7
$A \mathbin{\widehat{\cdot}} B$	$\alpha \cdot \beta$ via $\exp(\log \alpha + \log \beta)$	≈ 18
$A \mathbin{\widehat{/}} B$	α/β	≈ 22
$\widehat{\sqrt{}}(T)$	$\sqrt{\alpha}$ via $\exp(\frac{1}{2} \log \alpha)$	≈ 10
$\widehat{\mathrm{sq}}(T)$	α^2	≈ 14
$\widehat{\mathrm{inv}}(T)$	α^{-1} via $\exp(-\log \alpha)$	≈ 8
$T \mathbin{\widehat{\hat{}}} U$	α^β via $\exp(\beta \log \alpha)$	≈ 28

The complex variants $\widehat{\exp_{\mathbb{C}}}, \widehat{\log_{\mathbb{C}}}, \widehat{+_{\mathbb{C}}}, \widehat{-_{\mathbb{C}}}, \widehat{\cdot_{\mathbb{C}}}, \widehat{/_{\mathbb{C}}}$ have the same shape; their closure lemmas include extra side conditions on imaginary parts (the principal-branch strip).

Remark 7 (The shape of a closure lemma)**.** Every closure lemma has the form

$$\begin{aligned} \text{premise:} \quad & A \downarrow \alpha,\ B \downarrow \beta; \\ \text{side conditions:} \quad & \text{a fixed list of inequalities and} \\ & \text{non-vanishing conditions on } \alpha, \beta; \\ \text{conclusion:} \quad & A \mathbin{\widehat{\star}} B \downarrow \alpha \star \beta. \end{aligned}$$

The proof is mechanical: unfold the eml-tree from leaves outward, applying Definition 5 at each node and threading the side conditions to ensure that every $\log \circ \exp$ or $\exp \circ \log$ inverts. A tree of depth d produces a closure proof with roughly d "step" lines, each invoking one rule.

The addition combinator, made explicit. For concreteness, here is the smallest non-obvious case. Define

$$\begin{aligned} L_A &:= \mathrm{eml}\big(\mathbf{1},\ \mathrm{eml}(\mathrm{eml}(\mathbf{1}, \mathrm{eml}(A, \mathbf{1})), \mathbf{1})\big), \\ R_{A,B} &:= \mathrm{eml}\big(\mathrm{eml}(\mathrm{eml}(\mathbf{1}, \mathrm{eml}(A, \mathrm{eml}(A, \mathbf{1}))), \mathbf{1}),\ \mathrm{eml}(B, \mathbf{1})\big), \\ A \mathbin{\widehat{+}} B &:= \mathrm{eml}\big(L_A,\ \mathrm{eml}(R_{A,B}, \mathbf{1})\big). \end{aligned}$$

The closure proof unfolds $A \mathbin{\widehat{+}} B$ in eleven sequential steps, exhibiting the chain

$$\mathrm{eml}(A, \mathbf{1}) \downarrow e^\alpha, \qquad \mathrm{eml}(\mathbf{1}, \mathrm{eml}(A, \mathbf{1})) \downarrow e - \alpha, \qquad L_A \downarrow \alpha,$$
$$\mathrm{eml}(A, \mathrm{eml}(A, \mathbf{1})) \downarrow e^\alpha - \alpha, \qquad \mathrm{eml}(B, \mathbf{1}) \downarrow e^\beta, \qquad R_{A,B} \downarrow e^\alpha - \alpha - \beta,$$
$$\mathrm{eml}(R_{A,B}, \mathbf{1}) \downarrow e^{e^\alpha - \alpha - \beta}, \qquad A \mathbin{\widehat{+}} B \downarrow e^\alpha - \big(e^\alpha - \alpha - \beta\big) = \alpha + \beta.$$

The arithmetic identity at the heart of the construction is just $\alpha = e - \log(\exp(e - \alpha))$, the principal-branch involution applied twice.

5 The real-fragment compiler

With the combinator catalogue in hand, the compiler $\mathsf{EL} \to \mathsf{EML}$ is straightforward: a structural recursion on EL expressions in which each constructor is sent to its catalogue counterpart.

Definition 8 (Compiler)**.** The function compile : $\mathsf{EL} \to \mathsf{EML}$ is defined by recursion. On atoms,

$$\mathrm{compile}(\mathbf{1}) := \mathbf{1}, \qquad \mathrm{compile}(\mathbf{x}_n) := \mathbf{x}_n, \qquad \mathrm{compile}(\mathbf{0}) := \mathsf{Z};$$

on unary constructors,

$$\mathrm{compile}(\exp e) := \widehat{\exp}(\mathrm{compile}(e)), \qquad \mathrm{compile}(\log e) := \widehat{\log}(\mathrm{compile}(e)),$$

and similarly for the other unaries; on binaries, writing $\widetilde{e_i} := \mathrm{compile}(e_i)$ for brevity,

$$\mathrm{compile}(e_1 + e_2) := \widetilde{e_1} \mathbin{\widehat{+}} \widetilde{e_2}, \qquad \mathrm{compile}(e_1 \cdot e_2) := \widetilde{e_1} \mathbin{\widehat{\cdot}} \widetilde{e_2},$$

and similarly for $-$, $/$, ^, $\log_b$, avg, hypot. Here Z is a fixed closed witness for 0 — a small EML tree that evaluates to 0 unconditionally.

Theorem 9 (Compiler correctness)**.** *For every* $e \in \mathsf{EL}$*, every* $\mathrm{env} : \mathbb{N} \to \mathbb{R}$*, and every value* $v \in \mathbb{R}$*,*

$$e.\mathrm{eval?}(\mathrm{env}) = \mathrm{some}\, v \implies \mathrm{compile}(e).\mathrm{eval?}(\mathrm{env}) = \mathrm{some}\, v.$$

Sketch of the structure. By induction on e. Each constructor case dispatches to one closure lemma from §4: the induction hypothesis supplies the "$A \downarrow \alpha$" premise, the EL-level evaluation side conditions (e.g. $\alpha > 0$ for log, $\beta \neq 0$ for division) discharge the combinator's side conditions, and the conclusion is exactly what the closure lemma delivers. □

The architectural point is that the compiler's correctness proof contains *one combinator-application per constructor case*. All the heavy machinery (the eleven-step unfolding for $\widehat{+}$, the deep unfolding for $\widehat{\sqrt{\ }}$, etc.) is amortised inside the catalogue, proved once and then invoked.

6 The complex extension and Euler bridges

Why complex. Six paper primitives — π, i, cos, sin, tan, the inverse trig family — cannot be expressed in real EML alone. The underlying analytic identities (Euler's $e^{i\pi} = -1$, the exponential–trigonometric conversions) demand complex intermediates. We extend the grammar to $\mathcal{E}_{\mathbb{C}}$ — same syntax, complex evaluation — and copy the combinator catalogue with branch-cut side conditions.

Closed constants as building blocks. Five *closed* expressions appear repeatedly across the trig witnesses:

$$\mathsf{Z} \downarrow 0, \qquad \mathsf{C}_2 \downarrow 2, \qquad \mathsf{N}_i \downarrow -i, \qquad \mathsf{I} \downarrow i, \qquad \mathsf{P} \downarrow \pi.$$

Each is a finite $\mathcal{E}_{\mathbb{C}}$ tree paired with an unconditional evaluation lemma. Trig witnesses use them as plug-in factors rather than re-deriving Euler's formula at every step.

Real-to-complex lift. The embedding $\iota : \mathsf{EML} \to \mathsf{EML}_{\mathbb{C}}$ is a structural homomorphism: each constructor maps to itself, and only the evaluation rule changes. Its bridge lemma states: if $T.\mathrm{eval?}(\mathrm{env}) = \mathrm{some}\, v$ in the real semantics, then $\iota(T).\mathrm{eval?}(\mathrm{env}{\otimes}\mathbb{C}) = \mathrm{some}\,(v : \mathbb{C})$. This lets every complex trig witness import the *already-sealed* real arithmetic compositions ($\sqrt{1-x^2}$, x^2+1, $\frac{1}{2}$, etc.) without redoing branch-cut work.

The Euler bridges. Each trig witness encodes one Euler identity, then projects to the real or imaginary part. The full list:

$$\begin{aligned} \cos x &= \Re\big(\exp(ix)\big), \\ \sin x &= \Re\big(\exp(i(x-\tfrac{\pi}{2}))\big), \\ \tan x &= \Im\left(\frac{\exp(2ix)-1}{1+\exp(2ix)}\right) \quad \text{(Cayley quotient)}, \\ \arctan x &= \Im\big(\log(1+ix)\big), \\ \arccos x &= \Im\big(\log(x+i\sqrt{1-x^2})\big), \\ \arcsin x &= \Im\big(\log(\sqrt{1-x^2}+ix)\big). \end{aligned} \tag{1}$$

The right-hand side of each line is a finite $\mathcal{E}_{\mathbb{C}}$ witness once the catalogue combinators and the closed constants $\mathsf{I}, \mathsf{P}, \ldots$ are unfolded. The projection clause is then either $\Re$ or $\Im$, depending on which one carries the target real value.

7 Coverage and limits

The 36 paper primitives, by family.

- **Atoms (7).** Direct closed witnesses for $\mathbf{1}$, $\mathbf{x}_n$, and the named constants.
- **Real unaries (8) and binaries (8).** Sealed via Theorem 9: the $\mathsf{F36} \to \mathsf{EL}$ translation expands each primitive, the compiler then assembles a witness.
- **Hyperbolic family (6).** Same mechanism: sinh, cosh, tanh, arsinh, arcosh, artanh each translate to an EL composition of exp, log, $\sqrt{\ }$, and the four arithmetic operations.
- **Trig family (6).** Sealed via the Euler bridges (1), plus the closed-constant scaffolding of §6.

Three structural boundary points. The points $\sqrt{0}$, $\operatorname{arcosh}(1)$, $\operatorname{hypot}(0,0)$ all share the form *the natural EML witness collides with the junk-value boundary of* log. The paper itself does not provide explicit EML terms for these; we document them as out-of-scope boundary artefacts, with concrete counterexample evaluations.

Domain-narrowing companions. The Euler bridges as stated narrow at the principal-branch cut. To widen, we introduce *negative-side companions*: a witness T_f^- for the negative half-axis, constructed by feeding $-x$ — the real EL expression $\operatorname{neg}(\mathbf{x}_0)$ lifted along ι — into the same Euler-bridge skeleton with appropriately flipped constants. The two witnesses (positive and negative) seal a substantial open neighbourhood of 0, with $x=0$ itself filled in by a trivial constant witness.

8 A worked example: $\sin x$ end-to-end

We trace sin through every architectural layer, as the canonical case combining all the above.

Step 1 — source. The paper claim is

$$\exists\, \mathsf{T}_{\sin} \in \mathcal{E}_{\mathbb{C}},\ \forall x \in (0,\pi),\ \exists v \in \mathbb{C},\ \mathsf{T}_{\sin} \downarrow v\ \wedge\ \Re(v) = \sin(x).$$

Step 2 — choose the witness. From the Euler list (1), take the entry for sin and assemble its $\mathcal{E}_{\mathbb{C}}$ realisation:

$$\boxed{\mathsf{T}_{\sin} := \widehat{\exp_{\mathbb{C}}}\big(\,\widehat{\log_{\mathbb{C}}}(\mathsf{C}_{\cos})\ \widehat{-_{\mathbb{C}}}\ \widehat{\log_{\mathbb{C}}}(\mathsf{I})\,\big).}$$

Here $\mathsf{C}_{\cos}$ is the cosine-direction witness whose evaluation is e^{ix} for $x > 0$; it is itself a closed-constants $+\,\mathbf{x}_0$ assembly using I and $\widehat{+_{\mathbb{C}}}$ to encode $\log i + \log x = \log(ix)$ before two outer $\widehat{\exp_{\mathbb{C}}}$'s.

Step 3 — evaluation chain. Five sub-steps, each one closure-lemma application:

(i) $\mathsf{C}_{\cos} \downarrow e^{ix}$ (Euler bridge for e^{ix}, itself a small instance of the same template).

(ii) $\widehat{\log_{\mathbb{C}}}(\mathsf{C}_{\cos}) \downarrow ix$ (closure for $\widehat{\log_{\mathbb{C}}}$; side condition $\Im(ix) = x \in (-\pi, \pi]$ ✓).

(iii) $\widehat{\log_{\mathbb{C}}}(\mathsf{I}) \downarrow i\pi/2$ (closure for $\widehat{\log_{\mathbb{C}}}$, plus the Mathlib value $\log i = i\pi/2$).

(iv) $\widehat{\log_{\mathbb{C}}}(\mathsf{C}_{\cos})\ \widehat{-_{\mathbb{C}}}\ \widehat{\log_{\mathbb{C}}}(\mathsf{I}) \downarrow i(x - \pi/2)$ (closure for $\widehat{-_{\mathbb{C}}}$).

(v) $\mathsf{T}_{\sin} \downarrow \exp\big(i(x - \pi/2)\big)$ (closure for $\widehat{\exp_{\mathbb{C}}}$, unconditional).

Step 4 — projection. Take the real part:

$$\Re\Big(\exp(i(x - \tfrac{\pi}{2}))\Big) = \cos\Big(x - \tfrac{\pi}{2}\Big) = \sin x,$$

using the cofunction identity $\cos(x - \pi/2) = \sin x$. $\square$

Every layer of the architecture appears here in concentrated form: the grammar of §3, the catalogue of §4, the closed constant I and the Euler closed-constant scaffolding of §6, the real-to-complex lift implicit in any use of $\widehat{-_{\mathbb{C}}}$, the Euler identity for sin, and the principal-branch projection. The fact that the same template handles all six trig primitives — with only the inner content of C_f and the choice of projection $\Re$ or $\Im$ varying from line to line of (1) — is what gives the proof its uniformity.

9 What is novel

1. **The structural compiler.** Theorem 9 is a single induction with one combinator application per constructor. Each elementary composition reduces *uniformly* — no per-primitive cleverness. The heavy lifting (the eleven-step unfolding for $\widehat{+}$, the deep unfolding for $\widehat{\sqrt{\ }}$, etc.) is amortised inside the catalogue, proved once and re-used.

2. **Compositional branch-cut bookkeeping.** The closure lemmas for $\widehat{+_{\mathbb{C}}}$ and friends carry an explicit eleven-field precondition (the so-called *ADDsafe* bundle) that summarises all imaginary-part-in-strip and non-vanishing conditions the unfolding generates. Every trig witness's evaluation proof reduces to discharging this precondition by a few line-of-reasoning numerical inequalities. The branch-cut handling becomes *compositional*, not ad-hoc per primitive.

3. **The real-to-complex lift.** The homomorphism $\iota : \mathsf{EML} \to \mathsf{EML}_{\mathbb{C}}$ lets complex trig witnesses re-use real arithmetic that has already been structurally compiled. Without ι, every trig witness would have to reprove $\sqrt{1 - x^2}$ from first principles in the complex grammar; with it, the witness for arccos literally uses the real $\sqrt{1 - x^2}$ tree, lifted.

4. **Negative-side companions, from one observation.** A surprising amount of trig domain expansion (sin to $(-\pi, \pi)$, cos to $\mathbb{R} \setminus \{0\}$, arctan to $(-\pi, \pi) \setminus \{0\}$, etc.) follows from a single observation: the universal blocker is the same side condition (the strict inequality $\arg < \pi$ for $\widehat{\log_{\mathbb{C}}}$). Once $-x$ is encoded via ι, the same Euler-bridge skeleton runs on the flipped argument and produces the cofunction-shifted target. One template, five widenings.

10 The role of the principal branch

The fixed choice of branch for the complex logarithm is not a cosmetic detail of the construction; it is structural. Almost every quantitative constraint in the proof — the narrowed trig domains, the side-condition bookkeeping in the closure lemmas, the negative-side companion witnesses, and the structural exclusion of certain boundary points — traces back to the principal-branch convention. This section unpacks how.

The convention. Throughout the construction we use the principal branch

$$\log z \;=\; \log|z| \,+\, i\,\arg(z), \qquad \arg(z) \in (-\pi, \pi].$$

This is the convention of essentially every standard mathematical library. Its critical feature for our purposes is the inversion identity

$$\log(\exp w) \;=\; w \qquad \text{provided} \qquad \Im(w) \;\in\; (-\pi, \pi]. \tag{2}$$

Outside this strip, $\log \circ \exp$ wraps by integer multiples of $2\pi i$, breaking the identity.

How the constraint enters the closure lemmas. Recall the logarithm combinator

$$\widehat{\log}(T) \;:=\; \mathrm{eml}\big(\mathbf{1},\, \mathrm{eml}(\mathrm{eml}(\mathbf{1}, T),\, \mathbf{1})\big).$$

For $T \downarrow v$, the chain of eml-unfoldings produces intermediate values whose imaginary parts must lie in the strip (2) for the inversion to fire. The decisive step is the outermost $\mathrm{eml}(\mathbf{1}, \cdots)$, where one applies $\log\circ\exp$ to the value $e - \log v$, which has imaginary part

$$\Im(e - \log v) \;=\; -\,\Im(\log v) \;=\; -\,\arg(v).$$

The strip condition $\Im(\cdot) \in (-\pi, \pi]$ thus reads $-\arg(v) \in (-\pi, \pi]$, equivalently

$$\arg(v) \;\in\; [-\pi, \pi),$$

which excludes only one possible value: $\arg(v) = \pi$, the negative real axis. The closure lemma for $\widehat{\log}$ therefore carries the side condition

$$v \neq 0 \quad \text{and} \quad \arg(v) < \pi \text{ strictly}.$$

This is the core branch-cut constraint of the entire proof.

The cascade. Every higher combinator that internally applies $\widehat{\log}$ inherits this constraint. In particular:

- $\widehat{\cdot}_{\mathbb{C}}$ is implemented as $\widehat{\exp}\big(\widehat{\log_{\mathbb{C}}}(A) \mathbin{\widehat{+}_{\mathbb{C}}} \widehat{\log_{\mathbb{C}}}(B)\big)$, so it requires *both* $A.\!\downarrow$ and $B.\!\downarrow$ to satisfy the branch condition.
- $\widehat{/}_{\mathbb{C}}$ is implemented analogously and inherits the same constraints from each operand.
- Witnesses constructed by composition (the trig family, the inverse trig family, $\sqrt{\ }$ in the complex extension) carry a sum of constraints — one per occurrence of $\widehat{\log_{\mathbb{C}}}$ inside the witness.

This cascade is what determines the natural sealed subdomain of every trig primitive.

Concrete consequences. Three of the most visible narrowings have a direct branch-cut origin.

- sin *on* $(0, \pi)$. The witness applies $\widehat{\log_{\mathbb{C}}}$ to e^{ix}, whose argument is x; the side condition $\arg < \pi$ becomes $x < \pi$. Pushing past π would put $\arg(e^{ix})$ on the cut.
- arcsin *originally on* $(0, 1)$. The witness multiplies i by x via $\widehat{\cdot}_{\mathbb{C}}$, which requires $\arg(x) < \pi$; *positive* reals have $\arg = 0$ and pass, but *negative* reals have $\arg = \pi$ and fail. The bound $0 < x$ on the original witness is not a topological accident — it is exactly "x is not on the branch cut".
- tan *on* $(0, \pi/2)$. The Cayley quotient witness invokes $\widehat{\div}_{\mathbb{C}}$ on $2x$, again hitting the same constraint.

Branch-cut workarounds, made systematic. The widening companions of §7 are best understood as a unified collection of *branch-cut workarounds*, all leveraging the same observation: *if the natural witness fails because some sub-expression is a negative real, replace that sub-expression with* $-(-\bullet)$, *lifted into the complex grammar via* ι, *so the inner* $\bullet$ *is positive.*

For each of the five widenings, this reduces to choosing a different analytic identity:

- For arcsin, use $\arcsin x = \frac{\pi}{2} - \arccos x$. The witness for arccos already covers $(-1, 1)$ in full because its inner multiplication is $i \cdot \sqrt{1 - x^2}$, with $\sqrt{1 - x^2} > 0$ on the open interval, dodging the cut entirely.
- For arctan on the negative side, use $1 + ix = 1 - i \cdot (-x)$ with $-x > 0$.
- For cos, use the parity $\cos(-x) = \cos(x)$ to feed $-x$ into the same Euler skeleton.
- For sin, use $\sin x = \cos(\frac{\pi}{2} - x)$ together with $\log(-i) = -i\pi/2$ to keep the projection on the real part.
- For tan on the negative side, use the swap-numerator Cayley quotient $(1 - e^{-2ix})/(1 + e^{-2ix}) = i \tan x$.

Each identity is the analytic content; the branch cut is the technical constraint each identity is chosen to dodge.

An alternative would change the entire architecture. The paper itself sketches a different approach: redefine the branch of log inside the EML evaluator so that the cut runs elsewhere. Concretely, the paper's compiler "manually corrects" the sign of i when crossing the cut, effectively choosing a branch matched to the specific witness shapes used. With such a tailored branch, several trig primitives could be sealed on their full real domain by a single witness rather than by positive- and negative-side companions.

The price is structural: each closure lemma in the catalogue would have to be reproved against the alternative branch. Our construction chooses the standard convention (2) and pays the price in narrowed domains. Both choices are mathematically valid; they trade simplicity of side conditions for breadth of the sealed subdomain.

11 Open questions

The construction above resolves the headline existence claim of Theorem 1. Several structural and quantitative questions remain. We organise them into three groups: questions inherited from the paper's own list, questions about strengthening the present construction, and questions about the optimal expression sizes.

Questions from the paper

The paper's Supplementary Information lists seven open problems that neither the paper nor the present construction resolves. We restate the most accessible ones.

P1 Constant-free binary Sheffer. Does there exist a binary operator $\star : \mathbb{R}^2 \to \mathbb{R}$ (or $\mathbb{C}^2 \to \mathbb{C}$) such that $\{\mathbf{x}_n, \star\}$ alone, with no distinguished constant, generates all elementary functions? An exhaustive numerical search to operator complexity $K = 6$ has found no candidate, but no proof of impossibility is known.

P2 Real-only Sheffer. Does there exist a binary operator on $\mathbb{R}$ (no complex intermediates) that forms a complete basis for the elementary functions? The paper conjectures the answer is *no*: every candidate appears to require either a branch through $\mathbb{C}$ to compute π and the trigonometric family, or an extended-real intermediary $-\infty$. A formal impossibility proof is open.

P3 $-\infty$-free EML. Can the EML operator (or one of its cousins EDL, $-$EML) be used without the extended real axis, in particular without $-\infty$ as a possible intermediate value? In our partial-evaluation presentation we side-step this by returning $\perp$ instead of $-\infty$, but the question of whether the elementary functions are expressible *purely on the real line* is independent.

P4 Universal minimality of $\{1, \mathrm{eml}\}$. The pair (one constant, one binary operator) is empirically the smallest known calculator for the elementary functions. A truly universal minimality theorem would quantify over every continuous binary operator and every constant, asserting that no two-element basis is strictly smaller in any reasonable measure. The paper gives a cautionary trap example, $B(x, y) = x - y/2$, to illustrate why naive arguments fail.

P5 Stern–Brocot enumeration for EML expressions. The set of EML expressions is enumerable, but with massive redundancy: many syntactically distinct trees evaluate to the same elementary function. Is there a canonical form, analogous to the Stern–Brocot tree for rationals, that enumerates each elementary function exactly once?

P6 Taxonomy of the Sheffer family. The known cousins are EML, EDL ($\exp(x)/\log(y)$, requiring the constant e), and $-$EML ($\log(x) - \exp(y)$, requiring $-\infty$). Are these three points isolated, members of a discrete family, or random samples of a continuous one-parameter family of Sheffer operators?

Questions about the present construction

C1 Full-real-domain trig. Our construction seals the trig family on wide symmetric subdomains around 0, using positive- and negative-side companions. Can the same family be sealed on the *full* real domain by a single witness? Two paths suggest themselves:

- *Via a custom branch*: re-derive the catalogue against the paper's "manually corrected" branch. The closure lemmas change, but the witnesses gain wider sealed subdomains by construction.
- *Via multi-witness periodicity*: a witness family indexed by period number, with each member covering one fundamental period. The existential becomes weaker ($\forall x.\, \exists T$ rather than $\exists T.\, \forall x$), but the construction stays inside the principal branch.

Both are tractable but require non-trivial new infrastructure.

C2 Per-primitive completeness for the cousins. Theorem 1 concerns the EML operator. The paper presents EDL and $-$EML as confirmed-empirically-equivalent cousins; both deserve a parallel completeness theorem. The grammar scaffolding is straightforward (replace eml by the cousin operator); the witness search is genuinely new work, since the cousin operators have a different algebraic shape and the EML witnesses do not directly translate.

C3 The three structural boundary points. $\sqrt{0}$, $\operatorname{arcosh}(1)$, $\operatorname{hypot}(0,0)$. Each fails for the same reason: the natural EML expression for the value collides with the convention $\log(0) = 0$ that any total formalisation must adopt. Two candidate fixes:

- Extend the grammar with a primitive rpow operator that bypasses the log at the boundary.
- Move the affected witnesses entirely into the complex extension, where the boundary lives in different coordinates.

Neither has been pursued; both would extend the grammar beyond the paper's pure $\{1, \mathrm{eml}\}$ alphabet.

Quantitative questions on tree size

S1 Optimal tree size per primitive. Our construction produces witnesses whose tree sizes range from $K = 7$ (for 0) to $K \approx 10^7$ (for $\log_b$). The paper records small hand-tuned figures (e.g. $K = 5$ for sin via the identity $\sin x = \cos(x - \pi/2)$ in a setting where cos is already available). The gap reflects that our witnesses are produced by a uniform structural compiler, not optimised per primitive. What is the smallest K achievable for each primitive under the principal-branch convention?

S2 Asymptotic depth. The paper observes that the identity function appears at depth 4 in the EML tree, allowing variable transplanting by multiples of 4. Are there non-trivial functions whose minimum-depth witness has a different residue class modulo 4? If so, what depths are achievable?

S3 Compositional bounds. Given $K(f), K(g)$, what is the best upper bound on $K(f \circ g)$ derivable from the EML grammar? Our compiler achieves something like a multiplicative bound; tighter additive or near-additive bounds would substantially shrink the witnesses for composite primitives (and thereby narrow the gap with the paper's hand-tuned figures in S1).

12 Coverage scoreboard

Family	Sealed coverage	Mechanism
Atoms (7)	full	direct closed witness
Real unary (8)	open: full $\setminus\{\sqrt{0}\}$; boundary: witness-family	structural compiler + `GFullFix`
Real binary (8)	open: full $\setminus\{\operatorname{hypot}(0,0)\}$; boundary: witness-family	structural compiler + `GFullFix`
Hyperbolic (6)	open: full $\setminus\{\operatorname{arcosh}(1)\}$; boundary: witness-family	`F36` $\to$ `EL` + compiler + `GFullFix`
cos, arccos, arcsin (3)	full natural domain	Euler bridges + companions
sin, arctan, tan (3)	full natural domain (Path C′)	Euler bridges + range-reduction

Net. 36/36 paper primitives sealed by finite EML expressions on their natural domains. The three §G boundary points ($\sqrt{0}$, $\operatorname{arcosh} 1$, $\operatorname{hypot}(0,0)$) admit no *single* environment-independent witness because of Mathlib's $\log 0 = 0$ junk-value convention, but each is sealed by a *witness-family* theorem of the shape $\forall$env, [hyp] $\to \exists t \in \mathcal{E}_{\mathbb{C}}, t.\mathrm{eval}_? = \mathrm{some}\, v$ in the post-frontier module `GFullFix.lean` (§13). The trig family was narrowed to symmetric subdomains around 0 at submission time; the post-submission Path C′ widening (§13) lifts that narrowing for sin, arctan, tan. The headline theorem (Theorem 1) holds in its strongest form.

Public Lean API. 48 `paper_claim_*` theorems exported by `EML.Framework.PaperClaims`, plus 8 `edl_paper_claim_*` (Plan D) and 5 `negEml_paper_claim_*` (Plan E, including the `minusInf` witness via the parallel `NegEMLTermE` grammar over `EReal`) in `EML.Framework.Sheffer` — 61 original paper-claim theorems. The post-submission *frontier sprint* adds 39 more (§13), bringing the total to 100 **machine-checked theorems**. Local `lake build`: 8062 jobs, sorry-free. Every public theorem depends only on Mathlib's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`); see `VERIFICATION_EVIDENCE.md` for the verbatim `#print axioms` transcript. Independent re-verification on PCSS Eagle HPC: clean.

13 Addendum: post-submission progress (Path C′ + Plans A/D/E)

This section records four developments after the submission deck froze. The mathematical content of the original construction is unchanged; these are extensions, cleanups, and Sheffer-cousin progress.

Plan A — Sheffer naming cleanup

The submission-era `Sheffer.lean` contained four operator grammars (`EDLTerm`, `LDETerm`, `T1Term`, `T2Term`). Three of those names were misnomers relative to the paper's §3.1: `LDE` $= \log(x)/\exp(y)$ (division) was misnamed as the paper's $-$**EML** $= \log(x) - \exp(y)$ (subtraction); `T1Term` and `T2Term` were binary, but the paper's actual T_1 and T_2 are *ternary* operators (Supplementary Information §1.4) flagged by the paper itself as preliminary unverified candidates for a constant-free Sheffer.

After cleanup, `Sheffer.lean` hosts exactly the two paper-named cousins, EDL and $-$EML, with the correct definitions and a line-level paper-sourcing audit trail in `process_archive/legacy_planning/Sheffer_Pap`
The fabricated binary $\mathrm{T}_1/\mathrm{T}_2$ are removed; the paper's real ternary $\mathrm{T}_1/\mathrm{T}_2$ are documented as preliminary future work, not as part of the present scaffolding.

Plan B — architectural finding (custom log branch infeasible)

The paper at line 333 sketches an alternative route to full real-domain trig coverage by "manually correcting the i-sign" inside the compiler. Implementing this in Lean was the original Plan B and foundered on a structural fact: the $\mathsf{EML}_{\mathbb{C}}$ partial evaluation ($\mathtt{EMLTerm}^{\mathbb{C}}$`.eval?` in `Framework/Complex/Term.lean`) hard-codes Mathlib's principal-branch `Complex.log`; there is no place in the grammar to substitute a different branch convention.

The constructive observation — that $\mathtt{mkLog}_{\mathbb{C}}T$ *does* evaluate at the cut $\arg v = \pi$, returning $\mathrm{Complex.log}\, v - 2\pi i$ instead of $\mathrm{Complex.log}\, v$, so any witness whose final operation is $\exp_{\mathbb{C}}$ (such as cos) absorbs the $2\pi i$-shift via exp-periodicity — identifies why cos already covers $\mathbb{R} \setminus \{0\}$ but sin, arctan, tan do not. The fix-point of this analysis is that the paper's "manual correction" is necessarily a meta-level operation choosing a different witness shape per region of x, mathematically equivalent to the witness-family construction of Path C′.

Plan C′ — full-real-domain trig via range-reduction by substitution

A GPT Pro consultation (transcript in `process_archive/gpt_pro_bundle/trig_widening/`) gave a refined plan that uses the existing positive/negative companion witnesses but lifts them to full real domain by *substitution*. Three engineering moves:

1. **Variable substitution on $\mathcal{E}_{\mathbb{C}}$.** Define $\mathrm{subst}_0 : \mathcal{E}_{\mathbb{C}} \to \mathcal{E}_{\mathbb{C}} \to \mathcal{E}_{\mathbb{C}}$ that replaces every $\mathbf{x}_0$ in t with a given term s. Prove $\mathrm{eval}_?(t.\mathrm{subst}_0\, s, \mathrm{env}) = \mathrm{eval}_?(t, \mathrm{env}[\mathbf{x}_0 \mapsto \mathrm{eval}_?(s, \mathrm{env})])$. (`Subst.lean`.)

2. **Real-safe addition.** The lemma $\mathrm{ADDsafe}_{\mathbb{C}}((\overline{a}, 0), (\overline{b}, 0))$ — the gnarly 11-condition $\mathrm{mk}\widehat{+}_{\mathbb{C}}$ precondition bundle — holds automatically when both arguments are real-valued, the only nontrivial case being $\exp(a) - a \neq 0$ (true since $\exp(a) \geq a + 1$). With this foundation lemma, every period-shift via repeated $\mathrm{mk}\widehat{+}_{\mathbb{C}}$ stays in the real fragment, so the $\arg = \pi$ boundary trap never appears.

3. **Reuse, don't reshape.** For each narrow primitive, pick an algebraic identity that reduces to a primitive whose witness is *already* full-domain or full-natural-open:

   - $\sin x = \cos(\pi/2 - x)$: substitute $\pi/2 - x$ for $\mathbf{x}_0$ in the existing cos witness; finish with `Real.cos_pi_div_two_sub`. Coverage: $\mathbb{R} \setminus \{\pi/2\}$.
   - $\arctan x = \arcsin(x/\sqrt{1 + x^2})$: substitute the real-fragment compile of $x/\sqrt{1 + x^2}$ for $\mathbf{x}_0$ in arcsin's already-full witness; finish with `Real.arctan_eq_arcsin`. Coverage: full $\mathbb{R}$.
   - $\tan x = \tan(x - k\pi)$: range-reduce arbitrary x with $\cos x \neq 0$ to the fundamental strip $(-\pi/2, \pi/2)$ via repeated $\pm\pi$ shifts; apply the existing local Cayley witness on the reduced argument; finish with `Real.tan_sub_int_mul_pi`. Coverage: $\{x : \cos x \neq 0\}$.

Result. Three new public theorems $\mathrm{paper\_claim}_{\mathrm{sin\_full}}$, $\mathrm{paper\_claim}_{\mathrm{arctan\_full}}$, $\mathrm{paper\_claim}_{\mathrm{tan\_full}}$ — *witness-family* statements of the shape

$$\forall\, x \in \mathrm{Domain}(f),\ \exists\, t \in \mathcal{E}_{\mathbb{C}},\ \exists\, v \in \mathbb{C},\ t \downarrow v\ \wedge\ \pi(v) = f(x).$$

The witness depends on x's region (positive/negative side, or period number for tan); paper-faithfulness is preserved via the analogy with the paper compiler's manual sign correction.

Plan D — EDL per-primitive completeness

The paper's §3.1 promotes the EDL cousin $\mathrm{edl}(x, y) = \exp(x)/\log(y)$ to the same completeness status as EML, but supplies no Lean-style proof. Eight of the 36 paper primitives are now sealed in our framework via Aristotle-aided witness search:

Primitive	**Witness**	**Provenance**
1, $\mathbf{x}_0$, e	atomic	grammar
$\exp x$	$\mathrm{edl}(\mathbf{x}_0, e)$	identity (collapse)
$\log x$	$\mathrm{edl}(1, \mathrm{edl}(\mathrm{edl}(1, \mathbf{x}_0), e))$	Aristotle
x/y	$\mathrm{edl}(\mathrm{D8}(\mathbf{x}_0), \mathrm{D4}(\mathbf{x}_1))$	Aristotle
$\exp(\exp x)$	$\mathrm{edl}(\mathrm{edl}(\mathbf{x}_0, e), e)$	Aristotle
$\log(\log x)$	nested D8	Aristotle

Conditional ceiling scaffold. Aristotle's analytical work (transcripts in `process_archive/chunks/085_*`, `process_archive/chunks/089_*`) shows that the EDL combinator $\mathrm{edl}(a, b) = \exp(a)/\log(b)$ admits no mechanism for *addition* of sub-expression values. Multiplication ($x \cdot y = \exp(\log x + \log y)$), squaring, square-root, and the entire trigonometric and hyperbolic family are therefore conjecturally unreachable from closed EDL terms over $\{1, e, \mathrm{var}, \mathrm{edl}\}$. The constants -1, 2, $\frac{1}{2}$ are also conjecturally unreachable on Schanuel-conjecture grounds (their construction would require $\log 2 \in$ EL-closure of $\{1, e\}$, which Schanuel forbids). This validates the paper's own framing of EDL completeness as *conjectured*, not proven. The post-frontier module `EDLClosedVal.lean` formalises a *closed-value closure theorem* (proved unconditionally by induction on `EDLTerm`) and exposes three *conditional* obstruction corollaries (no closed EDL term evaluates to -1, 2, $\frac{1}{2}$), each guarded by a named typeclass `EDLTranscendenceBarrier` packaging the three non-membership facts a Schanuel-style argument would deliver. No instance of the typeclass is provided, so the corollaries are *scaffolded but not closed*; discharging them would require a Mathlib transcendence result that does not yet exist.

Plan E — −EML pilot

The third Sheffer cousin in the paper's §3.1 is $-$EML, paired with the constant $-\infty$. Two trivial atoms (1, var) are sealed in our $\mathbb{R}$-based `NegEMLTerm`. Sealing the $-\infty$ constant requires switching the grammar to EReal (extended reals); a self-contained pilot in `process_archive/chunks/088_neg_eml_pilot` demonstrates the approach. Per-primitive completeness for $-$EML is paper-open, with the same addition-unreachability obstruction as Plan D applying to the arithmetic and trig families.

Aristotle integration

Of the 10 chunks submitted in this round, 9 returned successful proofs, accounting for: two pure-Mathlib auxiliaries (`atanArg_in_Ioo`, `tan_period_reduction`); three end-of-pipeline assembly proofs for Path C$'$ (sin_via_cos, arctan_via_arcsin, tan_full); five Plan D / Plan E witness searches (D1–D11, E1–E3). Aristotle's role was strictly within the *audit blade* category: the architectural design came from human + GPT Pro, the framework integration was hand-coded, and Aristotle's contribution was witness search and proof-tactic discovery within self-contained chunk files.

14 Frontier sprint (2026-05-10/11)

After Path C$'$ closed the trig-widening gap, an independent GPT Pro consult (transcript in `process_archive/gpt_pro` identified four research-grade directions that the original artefact left open. A short focused sprint over two days delivered substantive Lean content for all four, adding 39 public theorems for a total of 100.

Direction 1 — SI §1.5 #5: variable-transplant identity depths

Module: `Framework/TransplantDepths.lean` (9 theorems).

The paper observes that the EML identity term has depth four and asks (SI §1.5 question 5) whether other identity-bearing depths exist. The module proves an *affirmative* family for every $d = 4k$ (an explicit term constructor parameterised by k) and *negative* closures for $d = 1$ and $d = 2$ (no EMLTerm of that depth is the identity on every real environment). Depth $d = 3$ is the smallest open case; the conjecture `NoIdentityAtDepthThree_conjecture : Prop` is stated and Aristotle returned a proof in a simplified single-atom grammar (`process_archive/chunks/090_*`); the canonical-grammar port is the natural follow-up.

Direction 2 — §G boundary points sealed via witness family

Modules: `Framework/GFullFix.lean` (3 theorems), `Framework/StructuralLimitsEReal.lean` (3 theorems).

The three boundary points ($\sqrt{0}$, $\operatorname{arcosh} 1$, $\operatorname{hypot}(0,0)$) admit no *single* witness because Mathlib's $\log 0 = 0$ makes the natural compositional witness fail. The fix is the same quantifier flip as Path C$'$: a *witness family* $\forall \text{env}, [\text{hyp}] \to \exists t, t.\text{eval}_? \,\text{env} = \text{some}\, v$, where on the boundary t is the constant-zero term mkZero and off-boundary t is the existing narrow paper-claim witness. The extended-real module `StructuralLimitsEReal.lean` confirms the boundary values independently in EReal arithmetic using ENNReal.log (which sends 0 to $\perp$ rather than 0, giving the mathematically correct limit).

Direction 3 — Plan D conditional-ceiling scaffold

Module: `Framework/EDLClosedVal.lean` (4 theorems + one typeclass).

Formalises the inductive predicate `EDLClosedVal` over $\mathbb{R}$ containing the values produced by closed `EDLTerm` evaluations, together with the closure theorem `edl_closed_eval_in_closedVal` proved unconditionally by induction on the term. The transcendence barrier (`EDLTranscendenceBarrier`) is packaged as a Lean `class` carrying three non-membership facts; three *conditional* obstruction corollaries (no closed EDL term evaluates to -1, 2, $\frac{1}{2}$) take the typeclass as a hypothesis. *No instance is provided*, so the corollaries are intentionally scaffolded rather than closed — a future Mathlib Schanuel-style result would discharge the typeclass and close them.

Direction 4 — Paper §5 universal-minimality, polynomial-class first cut

Module: `Framework/PolynomialBinary.lean` (2 theorems).

The paper asks whether the EML signature is minimal among single-binary-operator scaffolds (§5, line 533). The general question is open; the polynomial-class first cut is fully proved. A free term language `BTerm` over a generic binary B is introduced with predicate `IsPolynomialBinary` B asserting the existence of a bivariate polynomial witness $P \in \mathrm{MvPolynomial}(\mathrm{Fin}\,2, \mathbb{R})$ with $B\,x\,y = P\,[x, y]$. The *composition lemma* `polynomial_binary_terms_are_polynomial` shows that under a constant environment every `BTerm` reduces to a univariate polynomial (by induction on the term; the `app` case substitutes via `MvPolynomial.aeval`); the *impossibility theorem* `no_polynomial_binary_generates_exp` then contradicts the witness candidate with `Polynomial.tendsto_div_exp_atTop` (the polynomial-over-exp ratio tends to 0 at infinity, but under the candidate hypothesis it is constantly 1).

Direct-macro alternative witnesses (honest finding)

Module: `Framework/AlternativeWitnesses.lean` (9 paper-claim alternatives + 9 K-counts).

A parallel set of witnesses for the seven non-trivial binaries built directly from the unconditional macros in `Builders/Unconditional.lean`, bypassing the structural compiler. The honest finding (verified by `K_count_*_compact` theorems): direct-macro witnesses have *identical* K-counts to the compile-output ones, because the structural compiler already uses these macros internally. So the high K-counts of compile-output witnesses reflect the genuine cost of full-domain coverage, not a compiler inefficiency. The module was named `CompactWitnesses.lean` during initial development and renamed to `AlternativeWitnesses.lean` after the finding, because "compact" would have been misleading.

Sprint summary

Module	Theorems	Status
`TransplantDepths.lean`	9	sealed; $d = 3$ *Prop* stated
`StructuralLimitsEReal.lean`	3	sealed
`GFullFix.lean`	3	sealed
`EDLClosedVal.lean`	4 + class	closure sealed; corollaries conditional
`PolynomialBinary.lean`	2	sealed
`AlternativeWitnesses.lean`	$9 + 9$	sealed (honest finding)
Total	39	added to the original 61

The headline numbers after the sprint: 100 **public theorems**, 8062 **lake-build jobs**, 3 **§G boundary points sealed by witness families**, and an axiom audit showing every public theorem depends only on Mathlib's three standard axioms. The GPT Pro pre-announcement review (verdict: SHIP-WITH-FIXES; transcript in `process_archive/gpt_pro_bundle/pre_announcement_review/`) was executed on 2026-05-11 and its nine-item punch list completed in full before the public release.